

- 7 (a) (i) most α -particles were deviated through small angles B2 [2]
(allow 1 mark for 'straight through' / undeviated)
- (ii) small fraction of α -particles deviated through large angles M1
greater than 90° (allow rebound back) A1 [2]
- (b) e.g. β -particles have a range of energies
 β -particles deviated by (orbital) electrons
 β -particle has (very) small mass
(any two sensible suggestions, 1 each, max 2) B2 [2]

Do not allow β -particles have negative charge or β -particles have high speed